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Background Noise Assessment

Report No: 3410-R1 – South Marston, Swindon, – 23/11/2015

For: Brookbanks Consulting Ltd

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Report Revisions	*Date*		
3410-R1	23/11/2015	Initial Report	
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1. Introduction

A background survey has been carried out in order to assess the existing noise levels at a proposed development on an area of land to the east of Thornhill Road, South Marston, Swindon. The survey has been carried out on behalf of Brookbank Consulting Ltd to provide data for a noise boundary computer model. Existing background noise levels can be extrapolated to provide an indication of the likely affect of road noise on the proposed residential development.

The background survey was conducted at two monitoring locations covering a typical 24-hour period on Tuesday 10th November 2015.

2. Scope

Brookbank Consulting have been appointed as technical consultants in the development of a residential project at land to the east of Thornhill Road, South Marston, Swindon. The purpose of this report is to provide background noise levels representative of the normal condition to be found at positions bounding the proposed site. This will enable Brookbank Consulting to produce the required noise model as described in the introduction.

3. Site Description

The proposed residential scheme is to be developed on land to the east of Thornhill Road, South Marston, Swindon. The site is currently bounded to the north by South Marston Industrial Estate and agricultural land to other aspects. Further to the west of site is the Keypoint Rail Freight Terminal and Honda Car Factory.

An industrial unit immediately bounding the site identified as Synergy Health Laboratory has two extract flues on the roof which are clearly audible on an intermittent basis for short periods at the northern-western area of the proposed development. The flues operate intermittently during the daytime and night-time period.

Subjectively the background noise remains dominated by the road traffic noise from the surrounding area. Whilst attending site no activity noise from the KPRT and Honda factory was evident.

Figure 1 shows the proposed site layout and requested monitoring locations. The actual monitoring locations are illustrated in Figure 2, which were located on the site boundary.

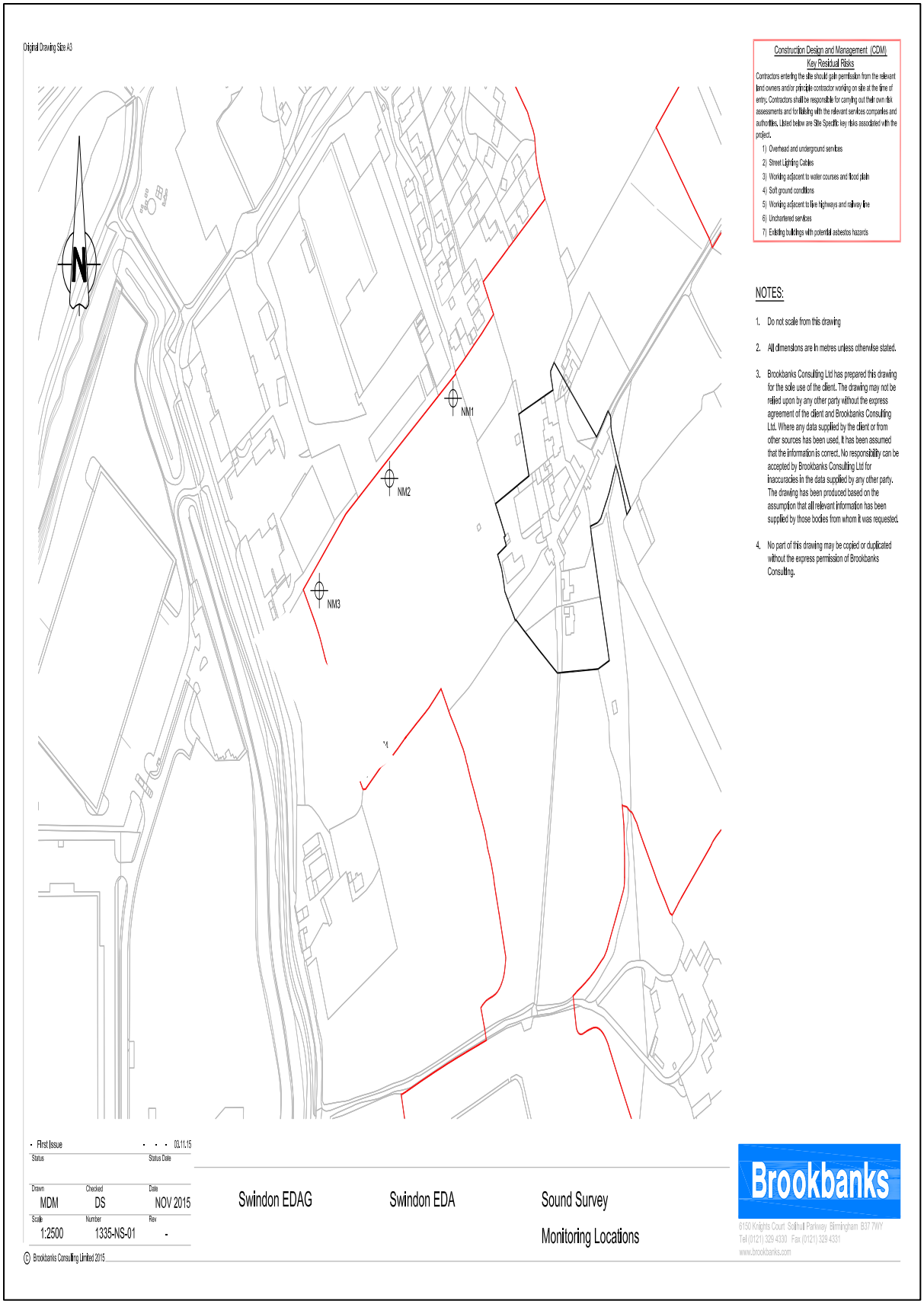

Figure 1 - Site Layout

4. Survey Information

Measurement Instrumentation

The measurement instrumentation used on the survey was as follows:

Equipment	Manufacturer & Type	Serial Number	Calibration Certificate
Sound Level Meter	Norsonic 118	28952	U12135
Sound Level Meter	Norsonic 118	30559	U10596
Sound Level Meter	Cassella CEL 480	089653	STD32308
Acoustic Calibrator	Norsonic 1251	32856	U12134

The equipment was calibrated to comply with section 4.2 of BS7445:1-2003 before and after the surveys. The calibration was as follows:

Meter	Serial	Before		After	
Norsonic 118	28952	113.9	-26.2	113.9	-26.1
Norsonic 118	30559	113.9	-26.3	113.9	-26.2
Cassella CEL 480	089653	114.0	-	114.0	-

Measurements & Timescales

During the background survey 5-minute measurements were made over a typical daytime and night-time periods on Tuesday 10th November 2015. The measurements were subsequently analysed into hourly periods.

During the attended survey 5-minute measurements were made to capture noise events from the extract flue and plant room of the adjacent industrial unit. The measurements were taken during the daytime of Friday 6th November 2015 (13:00-15:00) and Tuesday 10th November 2015 (00:00 – 01:00).

The following measurements are reported: L_{Aeq}, L_{A90}, L_{A10}, L_{AMax}

The measurements and their interpretation shall be in accordance with BS 7445: Parts 1 and 2. All sound pressure levels are in dB (re 20µPa).

Meteorology

During the monitoring the weather was considered to be dry with occasional light breeze.

Position of Monitoring Equipment

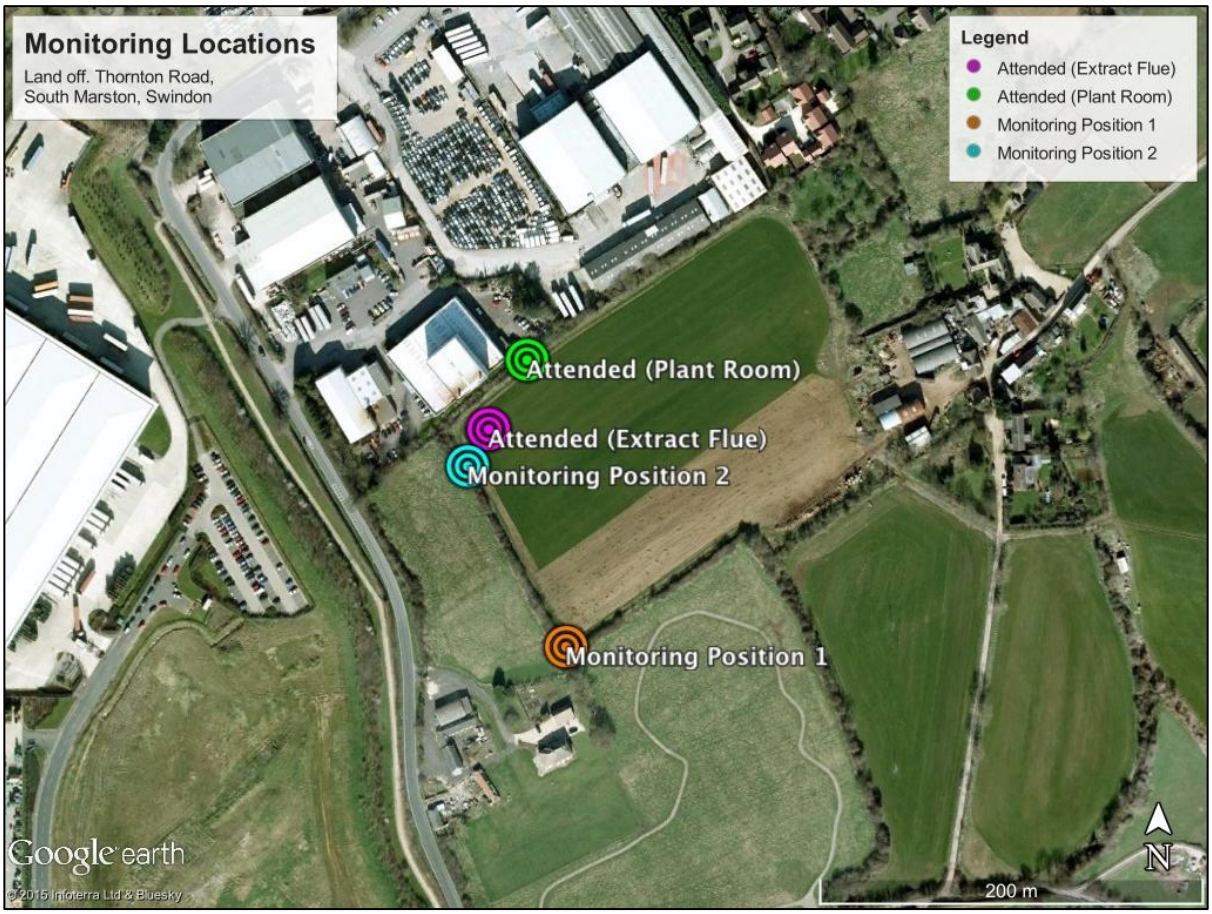

Figure 2 - Monitoring Locations

5. Survey Results

Background Noise Summary

The following tables show the summary of the background noise levels monitored.

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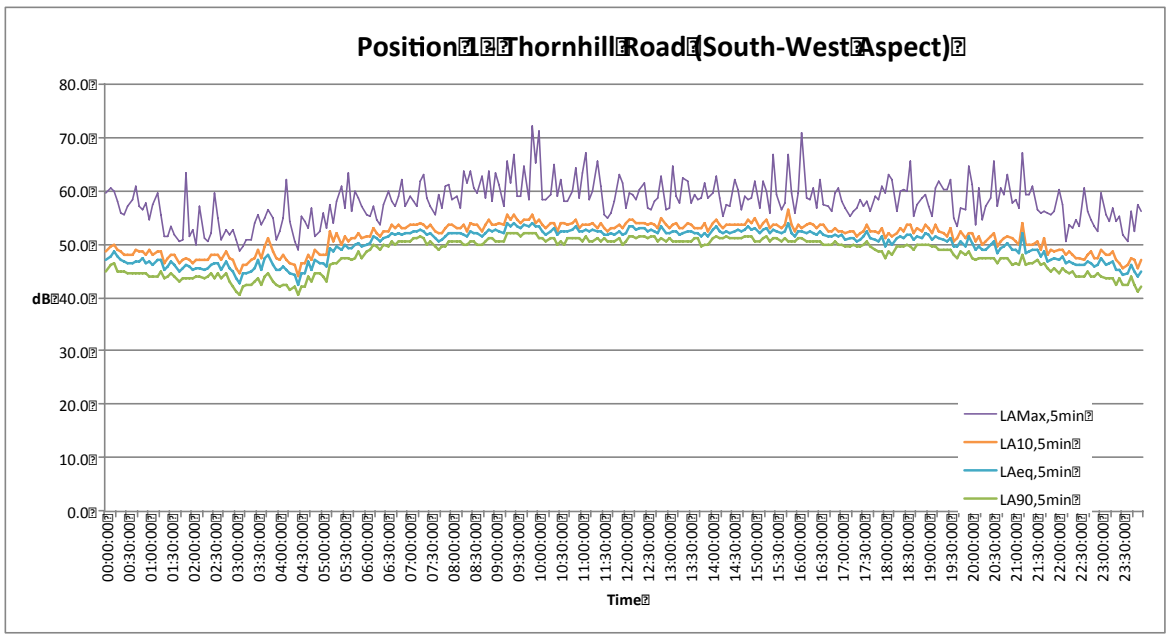
Site: Position 1 – Thornhill Road (South-West Aspect)
Table: 1 Summary — Start Date: 10/11/2015
Report: 3410-R1 — End Date: 10/11/2015

Hourly Summary				
Hour End	LAeq,1Hour	Average LA90,1Hour	Average LA10,1Hour	LAMax,1Hour
00:00	45.5	42.9	47.1	59.4
01:00	47.1	45.0	48.6	60.7
02:00	46.1	43.9	47.7	63.4
03:00	45.8	43.7	47.4	59.5
04:00	45.7	42.6	47.5	56.5
05:00	45.3	42.6	47.0	62.0
06:00	49.0	46.6	50.7	63.4
07:00	51.3	49.8	52.5	62.0
08:00	51.9	50.3	53.1	62.9
09:00	52.0	50.4	53.4	63.6
10:00	53.3	51.6	54.5	72.0
11:00	52.5	50.8	53.7	71.0
12:00	52.1	50.7	53.3	66.9
13:00	52.7	51.1	54.0	62.6
14:00	52.0	50.5	53.3	64.4
15:00	52.6	51.2	53.7	62.6
16:00	52.5	50.8	53.8	66.6
17:00	52.0	50.4	53.2	70.7
18:00	51.2	49.5	52.4	60.9
19:00	51.1	49.3	52.5	65.5
20:00	50.7	48.8	52.0	64.6
21:00	49.5	47.3	51.0	65.4
22:00	48.7	46.3	50.0	66.9
23:00	46.5	44.5	48.0	60.6



Monitoring Position 1 – Thornhill Road (South-West Aspect)				
Measurement	Daytime	Hour Ending/ Period	Night-time	Hour Ending/ Period
Maximum dB $L_{Aeq,1hr}$	53	10:00	51	07:00
Minimum dB $L_{Aeq,1hr}$	46	23:00	45	05:00
Average dB $L_{Aeq,16hr}$/$L_{Aeq,8hr}$	52	07:00 – 23:00	48	23:00 – 07:00

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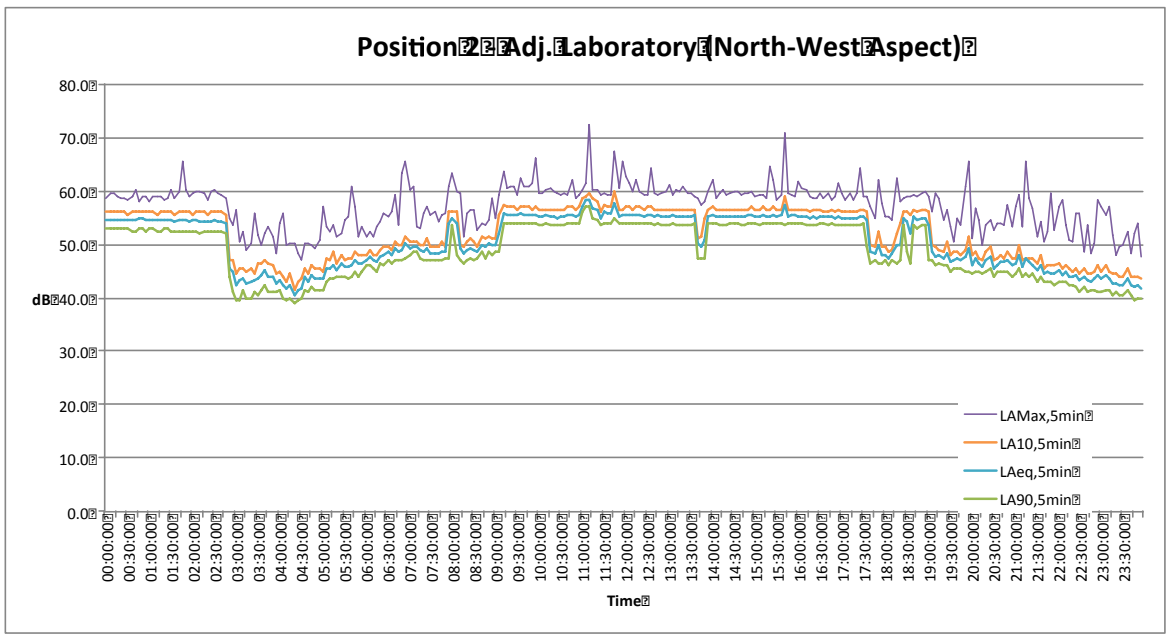
Site: Position 2 – Adj. Laboratory (North-West Aspect)
Table: 2 Summary
Report: 3410-R1
Start Date: 10/11/2015
End Date: 10/11/2015

Hourly Summary

Hour End	LAeq,1Hour	Average LA90,1Hour	Average LA10,1Hour	LAMax,1Hour
00:00	42.9	40.7	44.5	57.2
01:00	54.6	52.9	56.0	60.1
02:00	54.5	52.7	55.9	65.5
03:00	53.6	50.8	54.4	60.1
04:00	43.6	40.8	45.6	56.4
05:00	42.8	40.6	44.3	55.7
06:00	45.9	43.8	47.4	60.8
07:00	48.2	46.4	49.3	65.4
08:00	49.6	47.5	50.6	60.7
09:00	50.7	48.0	51.5	63.3
10:00	55.1	53.1	56.3	66.2
11:00	55.3	53.7	56.6	62.1
12:00	56.6	54.8	57.8	72.5
13:00	55.4	53.9	56.8	64.1
14:00	54.4	52.1	55.5	61.1
15:00	55.3	53.8	56.6	62.0
16:00	55.5	53.9	56.8	70.7
17:00	55.1	53.7	56.3	61.6
18:00	53.7	50.9	54.2	64.2
19:00	52.9	49.6	53.8	62.5
20:00	48.7	45.8	49.8	65.4
21:00	46.5	44.8	47.9	57.3
22:00	45.9	43.8	47.1	65.4
23:00	44.0	42.0	45.3	58.4

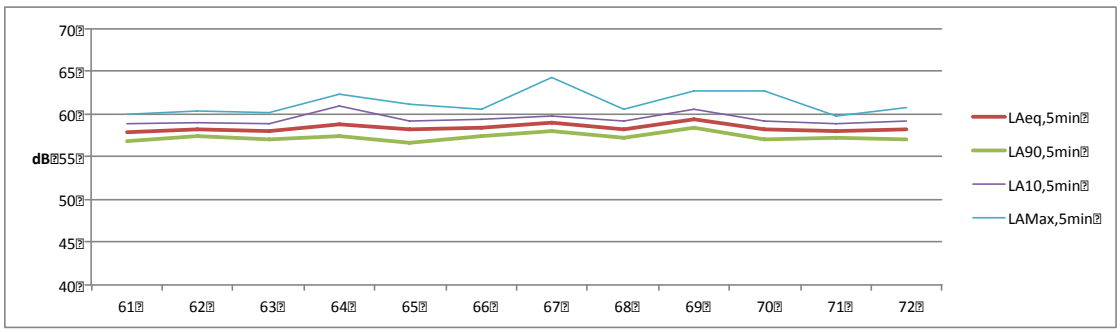


Monitoring Position 2 – Adj. Laboratory (North-West Aspect)				
Measurement	**Daytime**	**Hour Ending/ Period**	**Night-time**	**Hour Ending/ Period**
Maximum dB $L_{Aeq,1hr}$	57	12:00	55	01:00
Minimum dB $L_{Aeq,1hr}$	44	23:00	43	05:00
Average dB $L_{Aeq,16hr}$/$L_{Aeq,8hr}$	54	07:00 – 23:00	51	23:00 – 07:00

Site: Attended Monitoring South Marston
Table: Summary
Report: 3410-R1
Start Date: 06/11/2015
End Date: 06/11/2015

Attended Position (Extract System Daytime) Summary

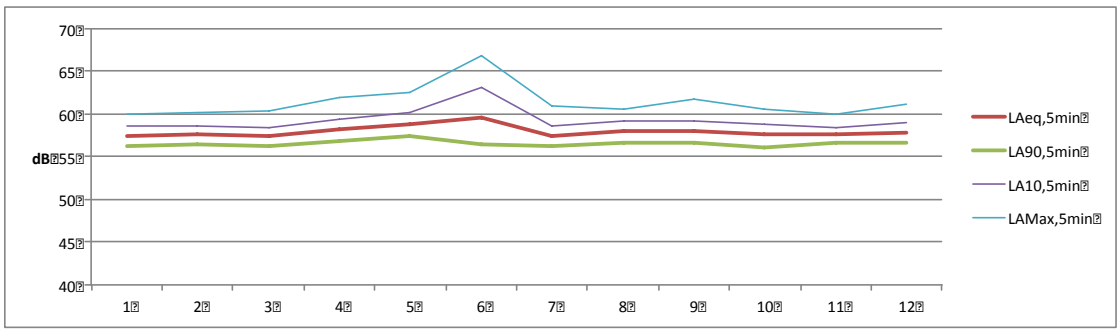
Run ID	LAeq,5min	LA90,5min	LA10,5min	LAMax,5min	Comments
61	57.9	56.9	58.9	60.0	
62	58.2	57.4	59.0	60.4	
63	58.0	57.0	58.9	60.2	
64	58.9	57.4	61.0	62.4	
65	58.2	56.6	59.3	61.1	
66	58.5	57.4	59.5	60.6	
67	59.0	58.1	59.7	64.3	
68	58.3	57.3	59.2	60.6	
69	59.5	58.4	60.5	62.7	
70	58.3	57.1	59.3	62.8	
71	58.1	57.3	58.9	59.7	
72	58.2	57.1	59.3	60.7	



Site: Attended Monitoring South Marston
Table: Summary
Report: 3410-R1
Start Date: 10/11/2015
End Date: 10/11/2015

Attended Position (Extract System Night-time) Summary

Run ID	LAeq,5min	LA90,5min	LA10,5min	LAMax,5min	Comments
1	57.4	56.3	58.6	59.9	
2	57.6	56.5	58.6	60.2	
3	57.4	56.3	58.4	60.3	
4	58.3	56.9	59.5	61.9	
5	58.8	57.5	60.1	62.5	
6	59.6	56.5	63.2	66.8	
7	57.5	56.2	58.6	60.9	
8	58.1	56.7	59.2	60.5	
9	58.0	56.7	59.3	61.7	
10	57.6	56.1	58.9	60.6	
11	57.6	56.6	58.4	59.9	
12	57.8	56.7	59.0	61.1	



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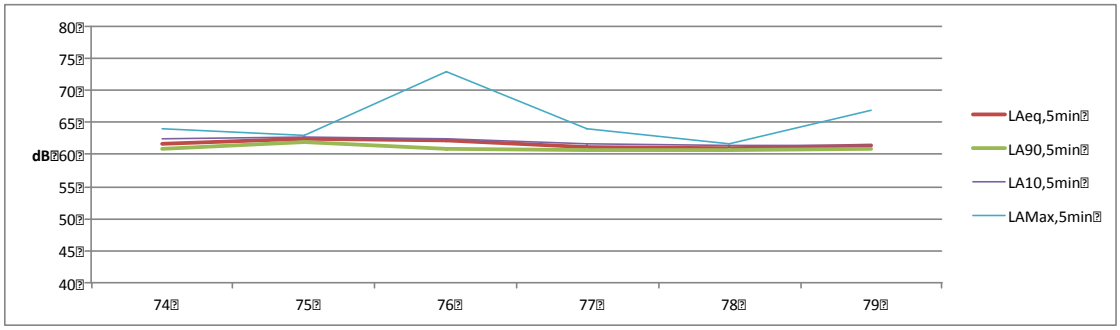
Site: Attended Monitoring South Marston
Table: Summary
Report: 3410-R1

Start Date: 06/11/2015
End Date: 06/11/2015

Attended Position 2 (Plant Room) Summary

Run ID	LAeq,5min	LA90,5min	LA10,5min	LAMax,5min	Comments
74	61.6	61.0	62.4	64.0	
75	62.4	62.0	62.8	63.1	
76	62.1	60.9	62.5	72.9	
77	61.2	60.7	61.8	64.1	
78	61.0	60.7	61.3	61.6	
79	61.5	60.8	61.4	66.8	



6. Conclusion

The site has been surveyed in accordance with client instructions in order to assess the current existing background noise levels and ambient levels from adjacent industrial estate over daytime and night time periods. The data collected can be used to contribute to the design of any mitigation measures which may be required for the planned development.

Steve Mckeever AMIOA

Acoustic Consultant

7. Appendix

Photo Appendix

Figure 3 – View of Proposed Development Site

Figure 4 – View of Synergy Health Care